

Summer of Science 2026

LARGE LANGUAGE MODELS

Name – Jagu Manish

Roll Number – 25M0612

Mentor – Ajay Kumar Meena

Objectives

- To understand the fundamentals of Artificial Intelligence, Machine Learning and Large Language Models (LLMs).
- To learn Python libraries and tools commonly used in AI applications.
- To study the working principles of transformers, embeddings, prompting and Retrieval Augmented Generation (RAG).
- To explore APIs and integration of LLMs into real-world applications.
- To build simple AI-based applications and chatbots using custom datasets.
- To understand the practical limitations, challenges and future scope of generative AI systems.
- To develop a final AI project and prepare technical documentation and presentation.

References

- Hands-On Large Language Models Language Understanding and Generation — Jay Alammar & Maarten Grootendorst
- FreeCodeCamp's Python Tutorial from YouTube
- Jay Alammar, *The Illustrated Transformer* — Understanding transformers, attention mechanisms and LLM workflow.
- Hugging Face LLM Course
- Hugging Face Transformers Documentation
- Prompt Engineering Guide — Study of zero-shot, few-shot and chain-of-thought prompting methods.
- LangChain Documentation — Building AI applications, chatbots and Retrieval Augmented Generation
- LlamaIndex Documentation — AI applications using custom datasets and document retrieval systems.
- freeCodeCamp AI Tutorials
- Andrej Karpathy Neural Networks and LLM Videos — Understanding neural networks and language models conceptually.
- Pinecone RAG Guide — Introduction to Retrieval Augmented Generation and vector search systems

Plan of Action

Week	Topics
May 24 - May 30 WEEK 1	• Basics of Python programming including functions, loops, dictionaries and file handling • Introduction to beginner-friendly libraries such as NumPy, Pandas and Matplotlib • Overview of Artificial Intelligence, Machine Learning, Deep Learning and Language AI
May 31 - June 6 WEEK 2	• Introduction to Large Language Models (LLMs) and their evolution • Understanding tokens, tokenization and embeddings • Conceptual understanding of transformer architecture and attention mechanisms • Understanding encoder and decoder models such as BERT and GPT
June 7 - June 13 WEEK 3	• Understanding how LLMs generate and process text • Prompt Engineering and effective prompting techniques • Study of zero-shot, few-shot and chain-of-thought prompting • Understanding limitations, hallucinations and ethical concerns in AI systems
June 14 - June 20 WEEK 4	• Working with APIs and LLM APIs using Python • Understanding REST APIs, JSON handling and API requests • Understanding the basic building of simple chatbot and AI applications using OpenAI and Hugging Face APIs • Introduction to Streamlit/Gradio interfaces
June 14 - June 20 Mid-Term Submission (June 3rd week)	Submission of progress report and overview of concepts learnt till mid-review
June 21- June 27 WEEK 5	• Development of mini project/final project based on LLMs and AI applications • Introduction to semantic search and Retrieval-Augmented Generation (RAG) • Basic understanding of vector databases and retrieval pipelines

June 28 - July 4	WEEK 6	• Understanding chatbot systems with memory and retrieval mechanisms • Introduction to LangChain and LlamaIndex frameworks • Understanding the development of AI applications using custom PDF/text data
July 5 - July 11	WEEK 7	• Testing, improving, and documenting the final project workflow • Exploring basic deployment and AI application workflows
July 12 - July 18	WEEK 8	• Preparation of final report, presentation, and project demonstration • Study of future scope, responsible AI usage, and real-world applications of LLMs. • Final revision of concepts and project documentation
July 12 - July 18	End-Term Submission (July 3rd week)	Submission of final report, presentation slides and project demonstration/video